MolSanity: A Reliability Audit for Feature Attributions on Molecular Graph Neural Networks

MolSanity contributors

[*author list and affiliations to be completed*]

https://github.com/Kar488/molsanity

Abstract. Feature attributions are routinely used to justify molecular graph neural network (GNN) predictions to chemists, yet they are almost never audited for *reliability*: existing evaluation frameworks ask whether an explanation is faithful to the model, not whether the model's explanation can be trusted, and not where it stops being trustworthy under the scaffold shift that characterises real drug-discovery workflows. We present **MolSanity**, a reliability-audit framework that wraps canonical attribution implementations (Captum, PyTorch Geometric, RDKit) rather than proposing a new attributor, and scores every (dataset × backbone × attributor × split) cell on six axes: motif-native coherence, occlusion–attribution faithfulness, ground-truth localisation where node labels exist, cross-checkpoint stability, calibration linkage, and confidence/correctness regime stratification. Our central finding is that *faithfulness is not correctness*, and that the gap opens under shift. On a synthetic task with *exact* node ground truth, holding the dataset, the backbone and the attributor set fixed and changing only the split, a faithfulness-only ranking is harmless in distribution — its top pick differs from the ground-truth-best attributor by 0.000 AUROC ($p = 0.109$) — but under scaffold shift all 3 of 3 ranking metrics, including our own occlusion measure, select an attributor that is significantly worse than the ground-truth-best (median gap 0.080 AUROC, paired Wilcoxon $p < 0.001$), and the faithfulness–truth rank correlation falls from 0.900 to 0.100. Across all 37 committed cells that carry a node-level ground truth, 13 are positively faithful to the model yet anti-aligned with the true motif. Reliability is not a property of the attributor alone: the backbone ordering does not survive the change of split ($\rho = -0.400$), and the best-performing, best-calibrated molecular cell in the matrix (BBBP · GCN, AUC 0.966, ECE 0.013) carries the most anti-faithful attributions of any ($\rho = -0.812$). Results are reported as of the committed `full.yaml` run, which completed 59 of 88 attempted cell-runs; the 27 failures and the 53 rows carried over from an earlier run are labelled as such throughout and never merged into a this-run aggregate. Every number, figure and table is regenerated from the committed artifacts by scripts in the repository.

Introduction

A medicinal chemist is shown a graph neural network's prediction that a candidate is hERG-active, together with a heat map over its atoms. The heat map concentrates on a piperidine nitrogen; the chemist finds that plausible, and the series is deprioritised. Nothing in that workflow ever established that the highlighted atoms are the atoms the model actually used, that they are the atoms that matter chemically, or that the highlighting behaves the same way on the next scaffold. Each of those is a separate question, and the answers come apart.

They come apart in a way that is measurable, and that gets worse in exactly the setting drug discovery operates in. An attribution can be *faithful* — the atoms it ranks highest really are the atoms whose removal moves the model's output most — and simultaneously *wrong* — those atoms are not the substructure that determines the property. Across the 37 committed cells in this study that carry a node-level ground truth, 13 are positively faithful to the model and yet *anti*-aligned with the true motif (ground-truth AUROC below the chance value of 0.5). A benchmark that scores only faithfulness rates all of them as fine.

The sharpest form of the problem is a selection problem. If you have no ground truth — which is the situation on every real molecular dataset in this study — you must choose an attributor using a faithfulness or fidelity metric. On a synthetic task where the ground truth *is* known exactly, we can check whether that choice is the right one, holding the dataset, the backbone and the attributor set fixed and changing only the split. In distribution the choice is harmless: the metrics' top pick is separated from the ground-truth-best attributor by 0.000 AUROC, which is not significant ($p = 0.109$, $n = 20$ molecules). Under a Bemis–Murcko scaffold split, all 3 ranking metrics — including MolSanity's own occlusion measure — select an attributor that is significantly worse than the ground-truth-best (median gap 0.080 AUROC, paired Wilcoxon $p < 0.001$), and the rank correlation between faithfulness and correctness falls from 0.900 to 0.100.

This is not an argument that faithfulness metrics are wrong. It is an argument that they answer a different question from the one a chemist is asking, and that the gap between the two questions widens under distribution shift. Molecular property prediction is deployed under scaffold shift almost by construction — the point of a screening model is to score series the training set does not contain — and the standard evaluation protocol for explanations does not stratify by that shift at all.

Contributions. This paper presents **MolSanity**, a reliability-audit framework for feature attributions on molecular GNNs. It is not a new attribution method: it wraps the canonical implementations (Captum [17], PyTorch Geometric [18], RDKit [19]) and audits them. Specifically:

- We define a six-axis audit over a motif-native (RDKit) decomposition: coherence, occlusion–attribution faithfulness, ground-truth localisation, cross-checkpoint stability, calibration linkage and confidence/correctness regime stratification, with field-standard Fidelity$\pm$, sparsity, the GraphFramEx characterisation score [10] and the PyG/DIG unfaithfulness metric [11] computed on the *same* molecules for direct comparability (The MolSanity audit).
- We show that a faithfulness-only ranking picks the wrong attributor under scaffold shift, in a design where only the split varies, so the effect is not confounded with a change of dataset (A fidelity-only ranking picks the wrong attributor under shift).
- We show reliability is a property of the cell, not of the attributor: the backbone ordering does not survive the change of split ($\rho = -0.400$ over 5 backbones), and faithfulness drops under shift in 17 of 29 cells audited on both splits (median $\Delta\rho = -0.106$).
- Using the 4253 per-molecule records this run committed, we stratify reliability by confidence/correctness regime and test the calibration-linkage hypothesis directly — finding it holds per cell (median $\rho = 0.166$, 20 cells significantly positive vs. 6 negative) but *reverses* when cells are pooled (-0.293), a Simpson's-paradox trap for anyone reporting a single aggregate number.
- We report the negatives with the positives: 27 of 88 attempted cell-runs failed in this run and are listed with their reasons; rows carried over from an earlier run are labelled and never merged into a this-run aggregate; and cells where the correctness axis simply does not exist are marked as such rather than scored.

Related work

Attribution methods. The attributors audited here are the standard ones. Gradient-family methods — Saliency [2], Input×Gradient, Guided Backpropagation [3] and Integrated Gradients [1] — are model-agnostic and cheap. Graph-specific methods learn a mask over the input: GNNExplainer [4] optimises a soft edge/feature mask per instance, PGExplainer [5] amortises that into a parametric mask network, and SubgraphX [6] searches connected subgraphs with Monte-Carlo tree search under a Shapley objective. Graph analogues of CAM/Grad-CAM were introduced by Pope et al. [7]; Yuan et al. [8] survey the space. We wrap, never reimplement.

Evaluating explanations. Two evaluation traditions exist. *Faithfulness* measures whether the explanation tracks the model: Fidelity$\pm$ and sparsity [8], the GraphFramEx characterisation score that combines them [10], and the unfaithfulness metric shipped with DIG/PyG [11]. *Correctness* measures whether the explanation recovers a known rationale, which requires ground-truth substructures — typically synthetic [9]. The tension between the two is known: Faber et al. [13] argue that comparing against ground truth is misleading when the trained model does not use that rationale, and Sanchez-Lengeling et al. [14] built the first systematic attribution benchmark for molecular GNNs on exactly this basis. Outside graphs, sanity checks [15] and retraining-based benchmarks [16] made a parallel case that plausible-looking attributions can be uninformative. MolSanity's position is that neither axis subsumes the other, and that the *joint* behaviour of the two under distribution shift is the object of interest.

Frameworks, and the gap. GraphXAI [9] supplies synthetic generators with exact ground truth and an evaluation library; GraphFramEx [10] systematises fidelity-style evaluation across explainers and tasks; DIG [11] packages methods and metrics; MolFaith [12] evaluates feature attributions on molecular property models under fidelity-based metrics. Table 1 places MolSanity against them. The claim is deliberately narrow: no individual row of that table is new. What is new is the combination — ground-truth localisation *and* faithfulness *and* cross-checkpoint stability *and* calibration linkage, computed over a motif-native decomposition, stratified by scaffold shift, across five backbones, with the field-standard metrics recomputed on the same molecules so the comparison is head-to-head rather than cross-paper.

Table 1: Where the audit sits relative to existing evaluation frameworks. ✓ = core capability, ∼ = partial or possible but not central, ✗ = not a focus. The contribution is the combination, not any single row: ground-truth validation and faithfulness both exist elsewhere, but not jointly with cross-checkpoint stability, calibration linkage and shift stratification over a motif-native decomposition.

Capability	GraphXAI	GraphFramEx	DIG	MolFaith	MolSanity
Ground-truth explanation benchmarks	✓ (synthetic)	∼	✓	✗	✓ (synthetic + motif-proxy)
Faithfulness / fidelity metrics	∼	✓	✓	✓	✓ (reproduced for comparability)
Molecular-motif-native audit (RDKit)	✗	✗	∼	∼	✓ (SSSR + Murcko + BRICS)
Occlusion–attribution agreement	∼	✓	∼	✓	✓
Cross-checkpoint stability	✗	✗	✗	✗	✓
Calibration linkage	✗	✗	✗	✗	✓
Scaffold-shift regime stratification	✗	∼	✗	✗	✓ (the core differentiator)
Multiple GNN backbones (agnostic)	∼	✓	✓	∼	✓ (GINE/GCN/GAT/MPNN/AttentiveFP)
Paired statistics (Wilcoxon, bootstrap CI)	∼	∼	∼	∼	✓
Wraps canonical implementations (no re-impl)	✓	✓	✓	✓	✓ (Captum/PyG/RDKit)

The MolSanity audit

Notation. A molecule is a graph $G = (V, E)$ with node features $X \in \mathbb{R}^{|V|\times d}$ and edge features. A model f maps G to a logit vector (classification) or a scalar (regression); t denotes the explained target. An attributor A produces node attributions $a = A(f, G, t) \in \mathbb{R}^{|V|}$. RDKit decomposes G into motifs $\mathcal{M} = \{M_1, \ldots, M_K\}$, $M_k \subseteq V$, from SSSR rings, the Bemis–Murcko scaffold [30] and fragment decomposition; the motif score is $s_k = \sum_{v\in M_k} \max(a_v, 0)$. Working at motif rather than atom granularity is what makes the audit chemically legible: a chemist reasons about a nitro group, not about atom 14.

Coherence. Does the attribution concentrate mass sensibly? We report the Gini coefficient of a, the mass in the top 20% of atoms, the largest connected-component fraction among salient atoms, and the motif top-1 share $\max_k s_k / \sum_k s_k$. Coherence is necessary but never sufficient: an attribution can be perfectly concentrated on the wrong motif.

Occlusion faithfulness. For each motif we zero its atoms' features through the model's node-mask multiplier and measure the drop in the target logit, $\Delta_k = f_t(G) - f_t(G \setminus M_k)$. Faithfulness is $\rho = \mathrm{Spearman}(s, \Delta)$ over the motifs of a molecule, plus the top-1 agreement indicator $\mathbb{1}[\arg\max_k s_k = \arg\max_k \Delta_k]$. **Higher ρ means more faithful**: the attributor ranks motifs the way occlusion does. $\rho < 0$ means the attributor ranks motifs in the reverse order of their causal effect on the model — worse than an uninformative explanation. For regression, Δ is taken in output space rather than probability space. We also compute Fidelity+ (target-probability drop when the salient atoms are removed; higher is better), Fidelity− (drop when the non-salient atoms are removed; lower is better), sparsity, the GraphFramEx characterisation score (their weighted harmonic mean [10]) and the PyG/DIG unfaithfulness metric [11], on the same molecules and the same masks.

Ground-truth localisation. Where a node-level ground-truth mask $g \in \{0, 1\}^{|V|}$ exists, correctness is the AUROC/AUPRC of the normalised a against g. Chance is 0.5; below 0.5 the attribution is anti-aligned. When g is degenerate (all-0 or all-1) the metric is undefined and reported as such — never as 0. Only the synthetic sets carry exact g; MUTAG carries a chemically motivated nitro/aromatic-nitro SMARTS proxy after Debnath et al. [28], and we label it a proxy everywhere. No molecular dataset in the sweep ships per-atom explanation labels, which is precisely why the synthetic anchor is needed.

Cross-checkpoint stability. An attribution that changes when the model is retrained a little is a weak basis for a chemistry decision. We keep an early checkpoint of the same run and report $\mathrm{Spearman}(s^{\mathrm{early}}, s^{\mathrm{final}})$ per molecule. This axis is absent from the frameworks in Table 1.

Calibration linkage and regime stratification. Predictions are temperature-scaled on validation [31] and we report test expected calibration error (ECE) alongside the attribution metrics. Per molecule, the audit assigns a regime from calibrated confidence and correctness — *confident-correct*, *confident-error* (the dangerous one) and *borderline*, at a confidence threshold of 0.8 — and stratifies the reliability metrics by it. Calibration linkage is the per-cell Spearman correlation between $1 - |\mathrm{confidence} - \mathrm{correct}|$ and a reliability metric: positive would mean better-calibrated predictions carry more trustworthy attributions.

Aggregation. Cells are compared with paired Wilcoxon signed-rank tests on shared molecules and bootstrap 95% confidence intervals over molecules. p-values are reported unadjusted and treated as descriptive.

Engineering. The pipeline is staged and idempotent: each stage writes a `.done` marker keyed to a content hash of its config, every cell is wrapped so a failure is logged and the run continues, and trained checkpoints are committed so the audit reproduces without retraining. Seeds, library versions, git revision and hardware are recorded in a per-run manifest.

Experimental setup

Datasets. Tier-1 provides the ground-truth anchor: **SynthMotifs** and its larger twin **SynthMotifsXL**, generated offline (Barabási–Albert base plus house/cycle motifs) with *exact* node ground truth, and **MUTAG** [28, 29] with the nitro proxy. Tier-2 is MoleculeNet [26]: **BBBP** and **BACE** (classification), **ESOL**, **FreeSolv** and **Lipophilicity** (regression). Tier-3 covers chemistry-meets-medicine endpoints: **ClinTox**, **SIDER** and **Tox21** as single-task binary views of the MoleculeNet panels, and **DILI** (drug-induced liver injury) and **hERG** (cardiotoxicity) via the Therapeutics Data Commons [27]. TDC sets are featurised with the same PyG encoding as MoleculeNet so the backbones are drop-in. BA-2Motifs and ShapeGGen [9] are attempted and gracefully skipped: the former's download returns HTTP 403 in this environment, the latter's install depends on the pre-2.5 PyG compiled extensions. Credential-gated resources are out of scope and never worked around.

Backbones. GINE [20, 21], GCN [22], GAT [23], MPNN [24] and AttentiveFP [25], sharing a pooling head, calibration and splits.

Attributors. Integrated Gradients, Saliency, Guided Backpropagation and Input×Gradient through Captum [17]; GNNExplainer and PGExplainer through PyTorch Geometric [18]. SubgraphX [6] requires DIG, which installs but cannot import in this environment (no `torch_sparse` wheel at the pinned torch version), so it is left blocked rather than approximated. PGExplainer is classification-only by construction: its mask network trains against class logits, so it is absent from the regression cells.

Splits. The primary split is a deterministic, class-aware Bemis–Murcko scaffold split [30] (whole scaffold groups move together, so there is no leakage, but every fold is guaranteed to contain every class), with a random split as the in-distribution reference. Imbalanced classification uses inverse-frequency class weighting and AUC-based model selection. Every comparison in this paper that is labelled "shift" versus "in-distribution" is these two splits on the same dataset.

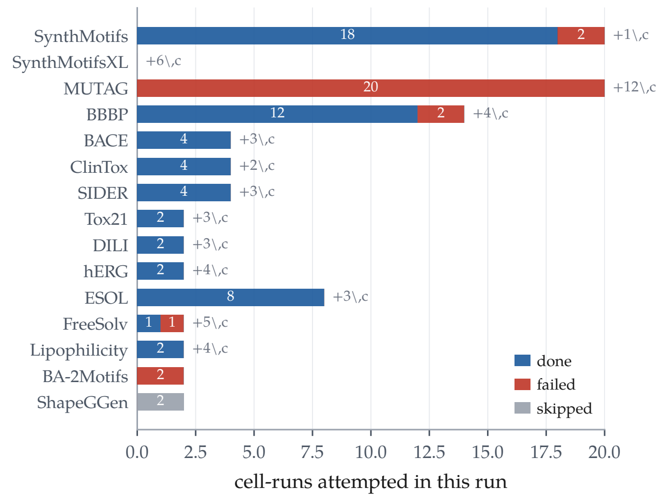

Figure 1: What the run actually produced. Outcome of every cell-run the `full.yaml` sweep attempted, by dataset. 27 of 88 failed — almost all of them to a device-placement defect that hit every MUTAG cell (20 of them) and both PGExplainer cells. "+n c" counts rows still present in the results matrix from an earlier reduced-budget CPU run for cells that failed here; those rows are marked ^{c} in every table and excluded from every this-run aggregate.

The run. All results are from the committed `full.yaml` sweep, run 20260730-164248 on a single CUDA device with seed 0 (torch 2.11.0+cu128, PyG 2.8.0.post1, RDKit 2026.03.4, Captum 0.9.0; manifest in `results/artifacts/run_manifest.json`). It attempted 88 cell-runs and committed 4253 per-molecule audit records across 59 cells.

Coverage, and what is carried. Table 2 and Figure 1 give the honest accounting: 59 cell-runs completed, 27 failed and 2 were skipped as unreachable. The failures group into 4 causes, dominated by two device-placement defects in the GPU path — 18 cells raised a CUDA-tensor-to-NumPy conversion error and 6 a device mismatch inside PGExplainer — with 1 batch-norm and 2 further failures. The consequence for this paper is specific and we do not paper over it: *every* MUTAG cell failed, so MUTAG contributes no per-molecule records to this run. Because the results matrix is keyed by cell and the last successful writer wins, 53 rows in it survive from an earlier reduced-budget CPU run. Those rows are real committed results, but they are not from this run, at this budget, on this hardware. We therefore mark them ^{c} everywhere, exclude them from every aggregate computed over per-molecule records, and keep the headline analysis (A fidelity-only ranking picks the

Table 2: The run ledger. Outcome of every cell the `full.yaml` sweep attempted, as recorded in `results/PROGRESS.md`. *carried* counts cells whose row still appears in the results matrix from an earlier reduced-budget CPU run because this run's attempt failed — those rows are marked [c] throughout this paper and are never mixed into a this-run aggregate.

dataset	done	failed	skipped	carried
BA-2Motifs	0	2	0	0
BACE	4	0	0	3
BBBP	12	2	0	4
ClinTox	4	0	0	2
DILI	2	0	0	3
ESOL	8	0	0	3
FreeSolv	1	1	0	5
Lipophilicity	2	0	0	4
MUTAG	0	20	0	12
SIDER	4	0	0	3
ShapeGGen	0	0	2	0
SynthMotifs	18	2	0	1
Tox21	2	0	0	3
hERG	2	0	0	4
total	**59**	**27**	**2**	**53**

Failure reasons. 18 × "can't convert cuda:0 device type tensor to numpy. Use Tensor.cpu()…"; 6 × "Expected all tensors to be on the same device, but got mat1 is on …"; 2 × "linear(): argument 'input' (position 1) must be Tensor, not NoneTy…"; 1 × "Expected more than 1 value per channel when training, got input si…".

wrong attributor under shift) entirely within cells this run produced. The matrix as committed contains 89 classification rows and 23 regression rows; 37 of them carry a node-level ground truth, 18 of those from this run.

Results

Faithfulness is not correctness

Table 3 lists every committed cell in which attribution *correctness* can be measured at all, and Figure 2 plots the joint distribution. Of the 37 cells with both metrics defined, 15 are anti-aligned with the ground truth (GT AUROC < 0.5) and 13 of those are simultaneously positively faithful to the model. That combination is the failure mode a faithfulness-only audit cannot see, because from the model's point of view nothing is wrong.

The clearest molecular instance is the MUTAG block of Table 3, where dataset, backbone and split are fixed and only the attributor varies: Saliency, Input×Gradient and Integrated Gradients have occlusion faithfulness 0.376, 0.398 and 0.414 — a spread of 0.038 — while their agreement with the nitro motif is 0.026, 0.042 and 0.540. Those rows are carried from the earlier run (this run's MUTAG cells all failed), and MUTAG's mask is a motif proxy rather than annotator labels, so we treat them as a motivating case study and put no weight on them in what follows. The load-bearing evidence is the next section, which uses exact ground truth and this run's records only.

A fidelity-only ranking picks the wrong attributor under shift

If faithfulness and correctness dissociate, then ranking attributors by a faithfulness metric — what an evaluation framework must do when no ground truth is available — can select the wrong method. The SynthMotifs · GINE cell family lets us test this cleanly: exact node ground truth, 5 attributors audited on the same molecules, and both splits available from this run, so the in-distribution and shift arms differ *only* in the split (Table 4, Figure 3).

In distribution, a fidelity-only ranking is adequate. Occlusion ρ selects GuidedBP, which is also the ground-truth-best attributor (1.000). Fidelity+ and the characterisation score both select Saliency instead, at GT AUROC 0.999 — a technical mismatch, but the two attributors are separated by 0.000 on paired per-molecule ground truth and the difference is not significant ($p = 0.109$, $n = 20$). Both sit at the ceiling. Rank correlations between metric and truth across the 5 attributors are 0.900, 0.400 and 0.700.

Under scaffold shift, the same experiment behaves differently. The ground truth ranks Saliency best (0.975), but all 3 metrics — occlusion ρ, Fidelity+ and characterisation alike — rank IG first, whose localisation is 0.901. On paired per-molecule GT AUROC over the 20 shared molecules the median gap is 0.080 with $p < 0.001$: the selected attributor is significantly worse, not merely different. The faithfulness–truth rank correlation drops to 0.100 for occlusion ρ and 0.400 for characterisation.

Two things are worth stating plainly. First, our own faithfulness metric fails here too. The conclusion is not that MolSanity's occlusion measure is a better selector than the field-standard ones; it is that under shift *no* faithfulness metric in this experiment carries reliable information about which attributor is correct, which is an argument for measuring correctness, not for a better proxy. Second, this experiment is on a synthetic, non-molecular graph task. It is the only setting in the sweep where exact ground truth, several attributors and both splits coexist, and that is precisely the point: on real molecular data the check we just ran cannot be run at all.

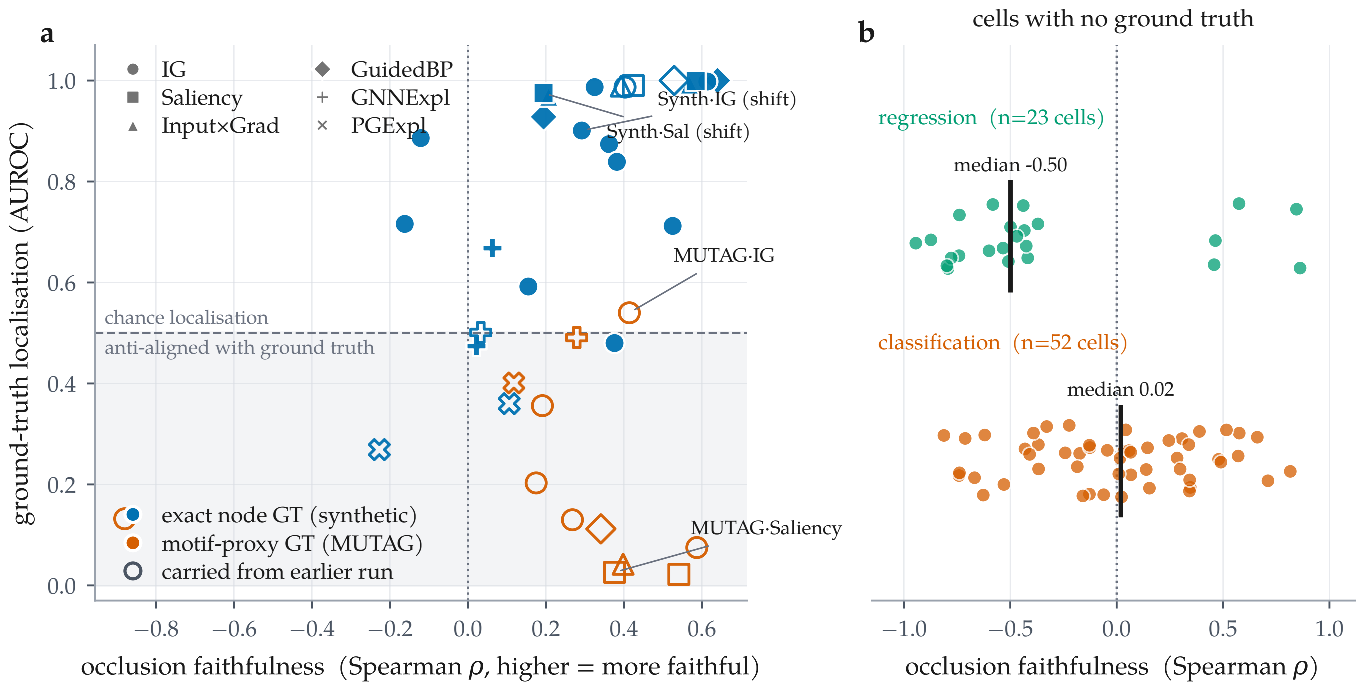

Figure 2: The two axes come apart, and for most cells only one of them exists. (**a**) Every committed cell carrying a node-level ground truth ($n = 37$), plotted as occlusion faithfulness against ground-truth localisation; colour distinguishes exact synthetic ground truth from the MUTAG motif proxy, marker distinguishes attributor, and open markers are rows carried from an earlier run. The shaded band is GT AUROC < 0.5: attributions there are anti-aligned with the true motif. 13 cells sit in the lower-right region — positively faithful to the model and anti-aligned with the truth. (**b**) The remaining 75 committed cells, on real molecular datasets, have no published per-atom labels: only the horizontal axis is measurable for them.

Reliability is a property of the cell

Figure 4 lays the two axes over the audited grid, and two further comparisons show that neither axis is a stable property of the attributor or of the architecture.

Backbones. Holding the attributor fixed at Integrated Gradients and varying the backbone (Figure 5) gives a spread of 0.518 AUROC in distribution (GINE 0.998 down to GCN 0.480) and 0.395 under shift (GCN 0.987 down to GAT 0.592) — comparable to the spread across attributors on the same dataset. The two orderings are negatively correlated ($\rho = -0.400$, 5 backbones): the architecture that localises the motif best in distribution is not the one that does so under shift. The attributor ordering is somewhat more robust across the same change of split ($\rho = 0.500$ over 5 attributors) but far from preserved.

The shift itself. Across the 29 cells this run audited on both splits, occlusion faithfulness falls under the scaffold split in 17 of them, with median change -0.106 (Figure 6b). Attributions do not merely become harder to validate under shift; they become measurably less faithful to the model that produced them.

Where attributions fail: regimes and calibration

Because this run committed its per-molecule records, two audit axes that previous drafts could only define are now measurable (Table 5, Figure 7).

Regime stratification. Pooling the 4253 audited molecules by regime, occlusion faithfulness is *negative* on confident-correct molecules (-0.218, $n = 1637$) and positive on borderline ones (0.259, $n = 1279$). Ground-truth localisation runs the other way, highest on confident-correct molecules (0.877, $n = 288$) and lower on borderline ones (0.655, $n = 70$). The confident-*error* regime — the one a deployment cares most about — has 109 molecules with a faithfulness value but only 2 with ground truth, so we report its GT mean in Table 5 and decline to interpret it. That scarcity is itself the finding: the regime where a wrong, confidently-made prediction gets a plausible explanation is the regime the current benchmark ecosystem provides almost no labelled molecules for.

Calibration linkage. The framework's hypothesis is that better-calibrated predictions carry more trustworthy attributions. Computed per cell it is supported on balance — median $\rho = 0.166$ across 48 classification cells, 20 significantly positive against 6 significantly negative

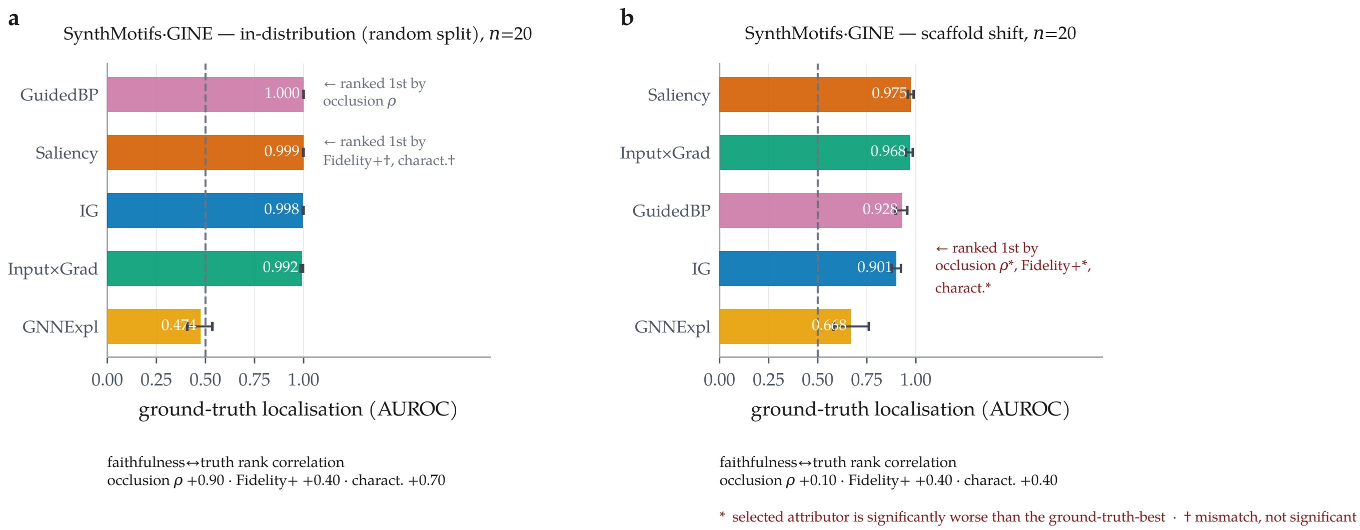

Figure 3: In distribution a faithfulness-only ranking is harmless; under scaffold shift it is not. Ground-truth localisation per attributor on the *same* dataset, backbone and attributor set — only the split differs — with bootstrap 95% confidence intervals over molecules; annotations mark which attributor each faithfulness-style ranking selects. (**a**) In-distribution: occlusion ρ selects the ground-truth-best attributor; Fidelity+ and the characterisation score select a different one whose GT AUROC is indistinguishable from the best ($p = 0.109$), marked †. (**b**) Under scaffold shift all three metrics select IG (GT AUROC 0.901) while the ground truth ranks Saliency best (0.975); the gap is significant on paired per-molecule tests, marked *. Rank correlations between each faithfulness metric and ground truth are given below each panel.

— but the effect is cell-specific rather than universal. Pooling every molecule into a single correlation instead yields -0.293: the opposite sign. The pooled number is an artefact of mixing cells with different confidence distributions, and we report it only to warn against it.

Cross-task reliability

Table 6 and Tables 8–9 give every committed cell on a real molecular dataset. Three cross-task patterns hold.

Regression is mostly, but not uniformly, anti-faithful. Across the 23 committed regression cells, 18 have negative occlusion faithfulness (median -0.499); the positive ones are ESOL·GAT, Lipophilicity·GAT, Lipophilicity·GINE. The same cells have well-behaved predictive performance (R^2 from 0.671 to 0.888 on ESOL). We do not interpret the sign, for a concrete reason visible in the table: the output-space Fidelity$\pm$ values reach 5.01, far outside the $[-1, 1]$ range a probability-space fidelity occupies, which shows the occlusion counterfactual is not calibrated for regression rather than showing that attributions there are inverted. The honest statement is that our regression faithfulness operator needs redefinition, and we flag it rather than build on it.

Coherence is high nearly everywhere, and says little. Motif top-1 share lies between 0.422 and 1.000 across the 87 molecular cells, with 73 of them inside a narrow 0.6–0.9 band, largely independent of whether the same cell is faithful or anti-faithful. Concentration on a single motif is easy to achieve and, on this evidence, close to uninformative about reliability. It belongs in the audit as a descriptive statistic, not as a quality score.

Stability is high but not uniform. Median cross-checkpoint stability over 59 cells is 0.832, with a minimum of -0.008 — one cell whose early and final attributions are essentially uncorrelated. By attributor, GNNExplainer is the most stable (median 0.950) and Integrated Gradients sits at 0.832 over 45 cells; the per-attributor counts are very unequal in this run, so these medians are descriptive rather than a controlled comparison.

Table 7 gives the paired attributor contrasts behind Section A fidelity-only ranking picks the wrong attributor under shift, computed from the per-molecule records.

Negative and degenerate results

We record what did not work, because it carries information.

A quarter of the run failed. 27 of 88 cell-runs failed (Table 2), overwhelmingly to two device-placement defects in the GPU code path rather than to anything scientific. The audit's per-cell isolation did its job — the run continued and committed 59 cells — but the cost is concentrated and material: MUTAG, the only molecular dataset with any ground-truth proxy, produced nothing this run.

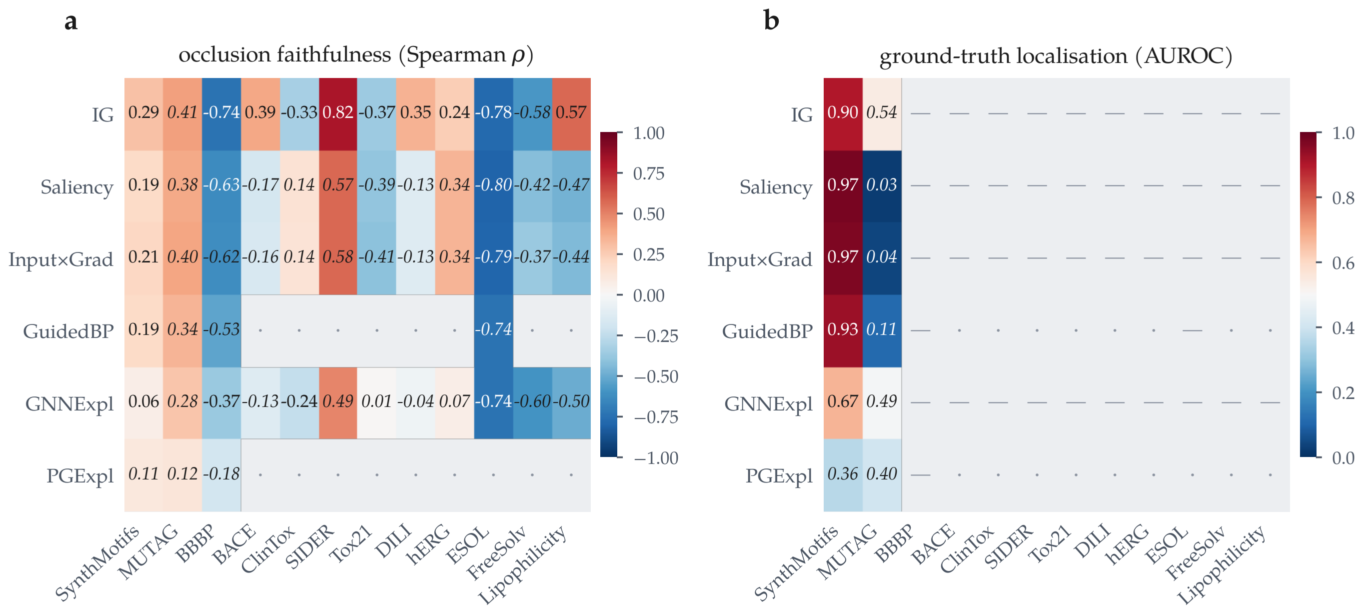

Figure 4: Reliability is a property of the (attributor, dataset) pair, not of the attributor. All committed GINE cells at the scaffold split. (**a**) Occlusion faithfulness; blue = anti-faithful. (**b**) Ground-truth localisation; the grey block is not a low score but an *unmeasurable* one — no molecular dataset here publishes per-atom labels. "—" marks an audited cell whose metric is undefined, "·" a cell that was not audited, and italic values are carried from an earlier run.

The best model has the worst attributions. The highest-AUC molecular classification cell in the run is BBBP · GCN under the scaffold split, at accuracy 0.980, ROC-AUC 0.966 and ECE 0.013 — accurate and well calibrated. Its attributions are the most *anti*-faithful of any molecular cell in the matrix ($\rho = -0.812$): Integrated Gradients ranks the motifs of these molecules in close to the reverse of their causal order. Predictive quality, calibration and explanation reliability are three different axes.

A weak model can be faithfully explained. The converse also appears. The most faithful molecular cell is SIDER · GINE ($\rho = 0.816$) on a model with accuracy 0.570 and ROC-AUC 0.650 — barely better than chance. Faithfulness certifies that the explanation describes the model; it says nothing about whether the model is worth describing. Any reliability report that omits the predictive metrics can be satisfied by this cell.

Accuracy is the wrong headline on imbalanced sets. 36 committed rows report test accuracy ≥ 0.95, with ROC-AUC as low as 0.805. On Tox21, accuracy runs 0.960–1.000 against AUC 0.822–0.849. We report AUC alongside accuracy throughout for this reason, and the lowest-AUC cell in the run (BACE · GCN, AUC 0.507 at accuracy 0.850) is exactly the case where accuracy alone would mislead.

Blocked, not silently dropped. SubgraphX (via DIG) and ShapeGGen (via GraphXAI) could not be installed: both need the pre-2.5 PyG `torch_sparse/torch_scatter` compiled extensions, for which no wheel exists at the pinned torch version. BA-2Motifs' download returns HTTP 403 here. All are logged as blocked, occupy no row in any table, and are excluded from every aggregate. ShapeGGen's exact-ground-truth role is covered by the offline SynthMotifs generator; SubgraphX's absence is a genuine gap in attributor coverage.

Discussion

What the audit buys. The single most useful thing MolSanity produces is not a score but a *negative certificate*: for 75 of the 112 committed rows — every real molecular dataset in the sweep — attribution correctness cannot be measured at all, because no per-atom explanation labels exist. Those cells can be audited for faithfulness, coherence, stability and calibration, and this paper's central result is that under shift none of those is a reliable stand-in for correctness. Making that gap explicit is more useful than filling it with a metric that looks like correctness and is not.

How attributions should be reported. Four practices follow directly from the matrix. (i) Report the regime: an attributor selected in distribution carries no warrant out of it, and on the one cell family where we can check, the selection is significantly wrong under shift. (ii) Report the backbone: the backbone ordering does not survive the change of split ($\rho = -0.400$), so "IG on a

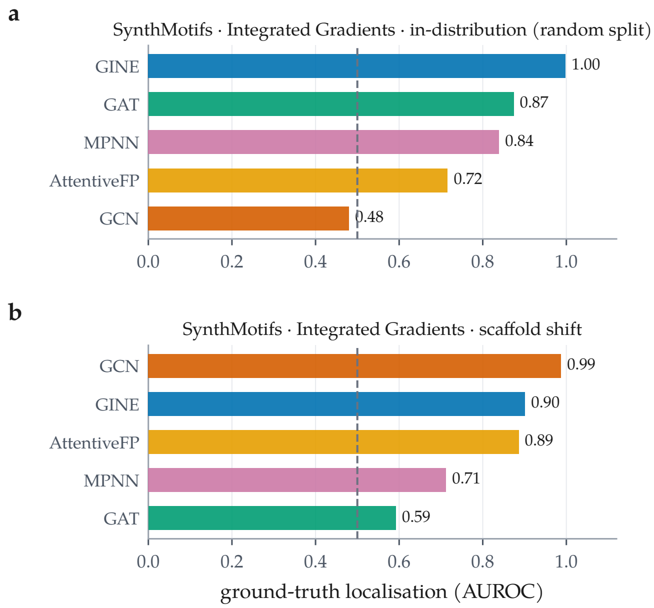

Figure 5: The backbone ordering does not survive the change of split. Integrated Gradients held fixed; ground-truth localisation across the 5 backbones on SynthMotifs, in distribution (**a**) and under scaffold shift (**b**). Dashed line is chance. Spearman correlation between the two orderings is −0.400 over 5 backbones.

GNN" is under-specified. (iii) Report the predictive metrics next to the explanation metrics, because the accurate, well-calibrated BBBP · GCN cell has the most anti-faithful attributions in the matrix. (iv) Do not pool: the calibration-linkage correlation changes sign between the per-cell and the pooled analysis.

Why reliability is regime-, backbone- and dataset-dependent. A faithfulness metric asks about the model; a ground-truth metric asks about the chemistry; they agree only when the model has learned the chemistry. That coincidence is engineered into synthetic benchmarks and is not guaranteed on molecular data, particularly out of distribution — the situation Faber et al. [13] identify as making ground-truth comparison hazardous in general. Attribution evaluation inherits the model's inductive limitations, which is why the correct object of study is the cell — (dataset, backbone, attributor, split) — and not the attributor.

Limitations and threats to validity

The headline experiment is synthetic. The clean within-dataset shift contrast exists only on SynthMotifs, a structural, non-molecular graph task with $n = 20$ audited molecules per arm. It validates the audit machinery and establishes the dissociation under a controlled shift; it does not establish chemical generalisation. Reproducing it on a molecular dataset requires per-atom labels that do not currently exist at scale, which is the same gap the paper is about.

A quarter of the grid failed, and MUTAG failed entirely. 27 of 88 cell-runs did not complete (Table 2). The failures are engineering defects, not scientific outcomes, but they shape what this draft can claim: the molecular-proxy ground-truth arm rests entirely on rows carried from an earlier, reduced-budget CPU run, marked ^{c} throughout. Those rows were produced at a different budget on different hardware and should not be compared numerically against this run's rows. All four defects have since been fixed in the repository, with regression tests (`tests/test_gpu_regressions.py`); recovering the affected cells requires a re-run, and this draft deliberately reports only what the committed run produced.

Single split, single seed. Each cell is one deterministic split at one seed. Cross-checkpoint stability partially probes run-to-run variability, but no confidence interval in this paper covers split or seed variance, and the per-cell molecule counts (20 on the synthetic arms) make individual contrasts underpowered.

Occlusion as a counterfactual. Zeroing node features takes the graph off the data manifold; a motif whose removal barely moves the output may be redundant rather than unimportant. This bears most heavily on the regression cells, where the output-space Fidelity± values leave the range a probability-space fidelity occupies and we decline to interpret the sign.

Proxy ground truth. MUTAG's mask is a nitro/aromatic-nitro SMARTS proxy, not annotator labels. It encodes an assumption about which substructure ought to matter, and the trained model may use a different, equally valid rationale [13].

Coverage of attributors. SubgraphX is absent for dependency reasons and PGExplainer failed in this run, so conclusions about the perturbation family rest on GNNExplainer alone in the current-run cells.

Multiplicity. p-values are unadjusted across a large number of paired comparisons; individual contrasts in Table 7 should be read as descriptive.

Conclusion

Molecular GNN attributions are audited today for whether they describe the model. Across the committed audit matrix we find that this is not the same question as whether they describe the chemistry, and that the two

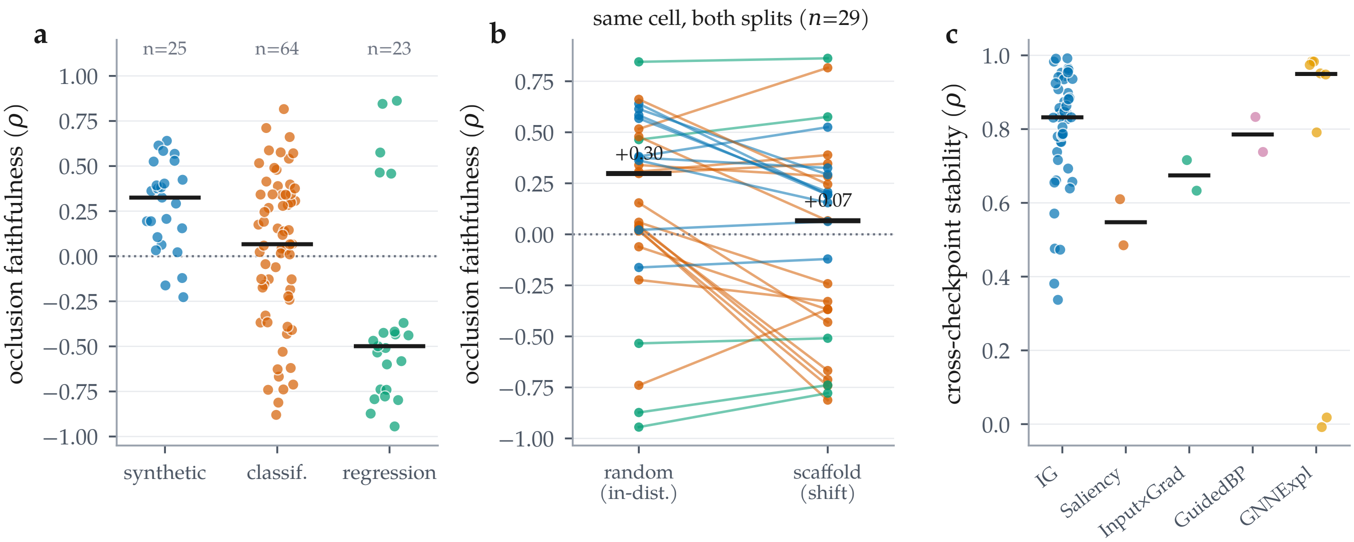

Figure 6: Distributions across the committed cells. (**a**) Occlusion faithfulness by dataset regime; bars are medians. (**b**) The same cell on both splits, joined: faithfulness falls under the scaffold split in 17 of 29 cells. (**c**) Cross-checkpoint stability by attributor; the mask-learning methods are not uniformly more stable than the gradient ones.

diverge in the regime molecular property prediction actually operates in. On the one cell family where exact ground truth, several attributors and both splits coexist, a faithfulness-only ranking is harmless in distribution and selects a significantly worse attributor under scaffold shift — and MolSanity's own faithfulness metric is no exception. Reliability does not transfer across backbones or across splits, and for the 75 molecular rows in the matrix, correctness cannot be measured at all with the labels the field currently has. The audit is what makes those failures visible; the pipeline, the figures and every number here regenerate from the committed artifacts as the grid fills in.

Data and code availability

All code, configurations, trained checkpoints, per-cell results, per-molecule audit records and the scripts that generate every figure and table in this paper are available at `https://github.com/Kar488/molsanity`. The committed run lives in `results/`: the audit matrix is `results/RESULTS.md`, the head-to-head comparison `results/BENCHMARK.md`, the run ledger `results/PROGRESS.md`, and the per-molecule records `results/artifacts/audit/`. This manuscript is built by `paper/figs/make_figures.py` and `paper/figs/make_tables.py`, which read those files directly; re-running them after a new run updates every number, figure and table in place. Dataset licences and provenance are recorded in the repository's dataset manifest.

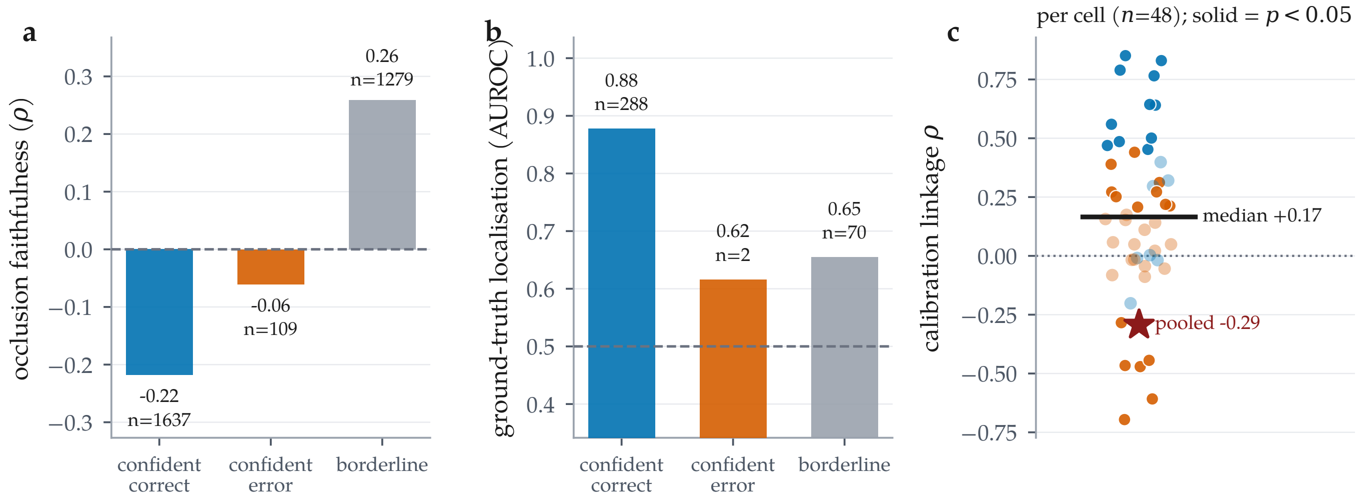

Figure 7: Regime stratification and calibration linkage, over the 4253 per-molecule records this run committed. (**a**) Occlusion faithfulness and (**b**) ground-truth localisation by confidence/correctness regime; n is printed per bar because ground truth exists for only a minority of molecules — the confident-error GT bar rests on 2 molecules and must not be read as an estimate. (**c**) Calibration linkage computed per cell (solid = $p < 0.05$): the median cell is positive, but pooling all molecules into one correlation reverses the sign — a Simpson's paradox, and a caution against reporting the pooled number.

Table 3: Every committed cell in which attribution *correctness* is measurable. Ground truth is exact for the synthetic sets and a chemically motivated nitro-motif proxy for MUTAG. GT AUROC = 0.5 is chance; below 0.5 the attribution is anti-aligned with the true motif. Occlusion ρ is the attribution–occlusion rank agreement (higher = more faithful to the model). Rows marked [c] are *carried*: they survive in the results matrix from an earlier reduced-budget CPU run because the corresponding cell failed in the latest run (Table 2); every other row was produced by the run in Table 2. Bold marks the best GT AUROC and the best occlusion ρ within each dataset×split block (emphasis only; values are unaltered).

dataset	backbone	attributor	split	n	acc	AUC	GT AUROC	GT AUPRC	occ. ρ	Fid+	Fid−	stab.
MUTAG[c]	GINE	IG	random	20	0.700	0.781	**0.075**	0.183	**0.587**	0.211	0.200	—
MUTAG[c]	GINE	Saliency	random	20	0.700	0.781	0.022	0.176	0.541	0.209	0.203	—
MUTAG[c]	AttentiveFP	IG	scaffold	20	0.650	0.967	0.132	0.150	−0.880	0.002	0.012	—
MUTAG[c]	GAT	IG	scaffold	20	0.750	0.890	0.130	0.151	0.268	0.189	0.247	—
MUTAG[c]	GCN	IG	scaffold	20	0.750	0.912	0.203	0.222	0.175	0.123	0.228	—
MUTAG[c]	GINE	GNNExpl.	scaffold	20	0.800	0.857	0.491	0.347	0.279	0.113	0.278	—
MUTAG[c]	GINE	GuidedBP	scaffold	20	0.800	0.857	0.112	0.144	0.341	0.272	0.229	—
MUTAG[c]	GINE	Input×Grad	scaffold	20	0.800	0.857	0.042	0.134	0.398	0.251	0.225	—
MUTAG[c]	GINE	IG	scaffold	20	0.800	0.857	**0.540**	0.405	**0.414**	0.252	0.245	—
MUTAG[c]	GINE	PGExpl.	scaffold	20	0.800	0.857	0.401	0.211	0.118	0.026	0.294	—
MUTAG[c]	GINE	Saliency	scaffold	20	0.800	0.857	0.026	0.131	0.376	0.267	0.230	—
MUTAG[c]	MPNN	IG	scaffold	20	0.750	0.868	0.356	0.274	0.191	0.263	0.312	—
SynthMotifs	AttentiveFP	IG	random	20	1.000	1.000	0.716	0.550	−0.162	0.335	0.296	0.655
SynthMotifs	GAT	IG	random	20	1.000	1.000	0.874	0.672	0.362	0.243	0.215	0.881
SynthMotifs	GCN	IG	random	20	0.600	1.000	0.480	0.339	0.376	0.119	0.142	0.693
SynthMotifs	GINE	GNNExpl.	random	20	1.000	1.000	0.474	0.250	0.022	0.171	0.222	0.018
SynthMotifs	GINE	GuidedBP	random	20	1.000	1.000	**1.000**	1.000	**0.640**	0.212	0.070	0.738
SynthMotifs	GINE	Input×Grad	random	20	1.000	1.000	0.992	0.968	0.568	0.212	0.084	0.633
SynthMotifs	GINE	IG	random	20	1.000	1.000	0.998	0.993	0.614	0.216	0.092	0.476
SynthMotifs	GINE	Saliency	random	20	1.000	1.000	0.999	0.997	0.584	0.234	0.068	0.485
SynthMotifs	MPNN	IG	random	20	1.000	1.000	0.839	0.668	0.382	0.410	0.459	0.473
SynthMotifs	AttentiveFP	IG	scaffold	20	0.850	0.970	0.886	0.624	−0.121	0.276	0.247	0.805
SynthMotifs	GAT	IG	scaffold	20	0.950	0.990	0.592	0.433	0.155	0.071	0.066	0.337
SynthMotifs	GCN	IG	scaffold	20	0.700	1.000	**0.987**	0.932	0.325	0.269	0.164	0.787
SynthMotifs	GINE	GNNExpl.	scaffold	20	0.950	1.000	0.668	0.360	0.063	0.201	0.168	−0.008
SynthMotifs	GINE	GuidedBP	scaffold	20	0.950	1.000	0.928	0.728	0.194	0.209	0.095	0.833
SynthMotifs	GINE	Input×Grad	scaffold	20	0.950	1.000	0.968	0.881	0.207	0.242	0.109	0.716
SynthMotifs	GINE	IG	scaffold	20	0.950	1.000	0.901	0.655	0.292	0.281	0.060	0.657
SynthMotifs[c]	GINE	PGExpl.	scaffold	20	1.000	1.000	0.360	0.220	0.106	0.064	0.126	—
SynthMotifs	GINE	Saliency	scaffold	20	0.950	1.000	0.975	0.915	0.194	0.256	0.099	0.610
SynthMotifs	MPNN	IG	scaffold	20	0.650	0.980	0.712	0.391	**0.525**	0.423	0.209	0.716
SynthMotifsXL[c]	GINE	GNNExpl.	random	120	1.000	1.000	0.501	0.259	0.033	0.364	0.472	—
SynthMotifsXL[c]	GINE	GuidedBP	random	120	1.000	1.000	**1.000**	0.998	**0.529**	0.495	0.006	—
SynthMotifsXL[c]	GINE	Input×Grad	random	120	1.000	1.000	0.989	0.958	0.393	0.493	0.089	—
SynthMotifsXL[c]	GINE	IG	random	120	1.000	1.000	0.987	0.945	0.403	0.493	0.021	—
SynthMotifsXL[c]	GINE	PGExpl.	random	120	1.000	1.000	0.269	0.165	−0.227	0.050	0.487	—
SynthMotifsXL[c]	GINE	Saliency	random	120	1.000	1.000	0.990	0.962	0.424	0.492	0.127	—

Table 4: Does a faithfulness-only ranking pick the ground-truth-best attributor? Both arms are the same dataset, the same backbone and the same 5 attributors — only the split changes — so the contrast isolates distribution shift rather than confounding it with a change of dataset. Each row ranks the attributors by one faithfulness/fidelity metric and asks whether its top choice is the one the exact node ground truth ranks best. p is a paired Wilcoxon test on per-molecule GT AUROC between the selected and the ground-truth-best attributor, over the molecules both audited. Recomputed here from the committed per-molecule records: the shipped `BENCHMARK_GT.md` is empty for this run because both of its target cells failed (Table 2).

regime	cell	ranking metric	its top pick	pick GT	GT-best	GT-best	mismatch	Wilcoxon p	ρ(faith,GT)
in-distribution	SynthMotifs·GINE, random, n=20	occlusion ρ	GuidedBP	1.000	GuidedBP	1.000	no	—	0.900
		Fidelity+	Saliency	0.999	GuidedBP	1.000	**yes**	0.109	0.400
		characterisation	Saliency	0.999	GuidedBP	1.000	**yes**	0.109	0.700
scaffold shift	SynthMotifs·GINE, scaffold, n=20	occlusion ρ	IG	0.901	Saliency	0.975	**yes**	$<$0.001	0.100
		Fidelity+	IG	0.901	Saliency	0.975	**yes**	$<$0.001	0.400
		characterisation	IG	0.901	Saliency	0.975	**yes**	$<$0.001	0.400

Table 5: Reliability stratified by confidence/correctness regime, pooled over the 4253 per-molecule records this run committed. A molecule is *confident-correct* or *confident-error* when its temperature-scaled confidence is at least 0.8, *borderline* otherwise. n is given per metric because ground truth exists for only a minority of molecules — in particular the confident-error GT mean rests on too few molecules to interpret, and is printed rather than hidden.

regime	occlusion ρ		GT AUROC		stability		motif top-1		confidence	
	mean	n	*mean*	n	*mean*	n	*mean*	n	*mean*	n
confident-correct	−0.218	1637	0.877	288	0.787	1603	0.684	1759	0.957	1759
confident-error	−0.061	109	0.616	2	0.846	108	0.782	114	0.909	114
borderline	0.259	1279	0.655	70	0.807	1273	0.724	1315	0.639	1315

Table 6: Molecular regression cells. Occlusion faithfulness is computed in output space (predicted-value shift) rather than probability space. In 18 of the 23 committed regression cells the attribution–occlusion agreement is *negative* — the atoms an attributor ranks highest are not the atoms whose removal moves the prediction most — while ESOL·GAT, Lipophilicity·GAT, Lipophilicity·GINE run positive. Note the Fidelity± columns leave the $[-1, 1]$ range a probability-space fidelity would occupy (up to 5.01), which is why we report but do not interpret their sign. Bold marks the highest occlusion ρ per dataset (emphasis only). PGExplainer is classification-only and therefore absent by construction.

dataset	backbone	attributor	split	n	RMSE	MAE	R^2	motif top-1	occ. ρ	occ. top-1	Fid+	stab.
ESOL	GAT	IG	random	100	0.730	0.549	0.888	0.850	0.845	0.970	3.52	0.908
ESOL	GAT	IG	scaffold	100	0.793	0.627	0.838	0.838	**0.862**	0.960	5.01	0.935
ESOL	GCN	IG	random	100	0.951	0.746	0.809	0.870	−0.534	0.330	−0.83	0.936
ESOL	GCN	IG	scaffold	100	1.017	0.793	0.734	0.870	−0.509	0.350	−0.88	0.965
ESOL	GINE	GNNExpl.	random	100	0.788	0.599	0.869	0.868	−0.873	0.230	−1.35	0.948
ESOL	GINE	GNNExpl.	scaffold	100	0.929	0.724	0.778	0.853	−0.739	0.280	−0.91	0.974
ESOL [c]	GINE	GuidedBP	scaffold	20	1.132	0.918	0.671	0.659	−0.741	0.150	−0.58	—
ESOL [c]	GINE	Input×Grad	scaffold	20	1.132	0.918	0.671	0.881	−0.793	0.100	−1.02	—
ESOL	GINE	IG	random	100	0.788	0.599	0.869	0.878	−0.944	0.230	−1.73	0.960
ESOL	GINE	IG	scaffold	100	0.929	0.724	0.778	0.878	−0.778	0.290	−1.39	0.954
ESOL [c]	GINE	Saliency	scaffold	20	1.132	0.918	0.671	0.878	−0.798	0.100	−1.00	—
FreeSolv [c]	GCN	IG	scaffold	20	2.567	2.048	0.538	0.782	−0.434	0.400	−0.05	—
FreeSolv [c]	GINE	GNNExpl.	scaffold	20	2.347	1.733	0.614	0.766	−0.600	0.350	−0.42	—
FreeSolv [c]	GINE	Input×Grad	scaffold	20	2.347	1.733	0.614	0.805	**−0.370**	0.450	−0.36	—
FreeSolv	GINE	IG	random	65	1.486	1.081	0.803	0.864	−0.418	0.523	−0.66	0.847
FreeSolv [c]	GINE	IG	scaffold	20	2.347	1.733	0.614	0.782	−0.582	0.400	−0.54	—
FreeSolv [c]	GINE	Saliency	scaffold	20	2.347	1.733	0.614	0.789	−0.425	0.450	−0.37	—
Lipophilicity [c]	GAT	IG	scaffold	20	0.940	0.746	0.391	0.762	0.458	0.900	−0.14	—
Lipophilicity [c]	GINE	GNNExpl.	scaffold	20	0.932	0.740	0.402	0.845	−0.499	0.000	−0.25	—
Lipophilicity [c]	GINE	Input×Grad	scaffold	20	0.932	0.740	0.402	0.699	−0.439	0.000	−0.28	—
Lipophilicity	GINE	IG	random	100	0.737	0.571	0.617	0.775	0.464	0.720	0.52	0.832
Lipophilicity	GINE	IG	scaffold	100	0.749	0.584	0.614	0.760	**0.575**	0.860	1.03	0.863
Lipophilicity [c]	GINE	Saliency	scaffold	20	0.932	0.740	0.402	0.705	−0.469	0.000	−0.30	—

Table 7: Paired attributor comparisons on shared molecules, computed from the committed per-molecule records (Wilcoxon signed-rank on per-molecule occlusion faithfulness, $\Delta = A - B$). Same cell family as Table 4. p values are unadjusted for multiplicity and should be read as descriptive at these sample sizes.

A vs. B	n	**median** Δ	p
SynthMotifs, GINE, random split			
IG vs. Saliency	20	0.043	0.277
IG vs. Input×Grad	20	0.087	0.231
IG vs. GuidedBP	20	0.120	1.000
IG vs. GNNExpl.	20	0.635	<0.001
Saliency vs. Input×Grad	20	0.030	0.189
Saliency vs. GuidedBP	20	0.022	0.622
Saliency vs. GNNExpl.	20	0.588	<0.001
Input×Grad vs. GuidedBP	20	−0.020	0.189
Input×Grad vs. GNNExpl.	20	0.558	<0.001
GuidedBP vs. GNNExpl.	20	0.684	<0.001
SynthMotifs, GINE, scaffold split			
IG vs. Saliency	20	0.091	0.294
IG vs. Input×Grad	20	0.128	0.409
IG vs. GuidedBP	20	0.153	0.294
IG vs. GNNExpl.	20	0.212	0.001
Saliency vs. Input×Grad	20	−0.033	0.083
Saliency vs. GuidedBP	20	−0.045	0.784
Saliency vs. GNNExpl.	20	0.109	0.114
Input×Grad vs. GuidedBP	20	−0.014	0.812
Input×Grad vs. GNNExpl.	20	0.097	0.123
GuidedBP vs. GNNExpl.	20	0.243	0.189

Table 8: Molecular classification cells (no node-level ground truth available), 1 of 2. Every committed classification cell on a real molecular dataset. The ground-truth localisation column does not exist for these datasets — no per-atom labels are published — so only model-side reliability can be measured, which is precisely the gap the Tier-1 cells are needed to close. *charact.* is the GraphFramEx characterisation score and *unfaith.* the PyG/DIG unfaithfulness metric, computed on the same molecules; both are blank for carried rows ([c]), whose per-molecule records this run did not regenerate. Bold marks the highest occlusion ρ per dataset (emphasis only).

dataset	backbone	attributor	split	n	acc	AUC	ECE	motif top-1	occ. ρ	occ. top-1	Fid+	Fid−	stab.	charact.	unfaith.
BACE	GCN	IG	random	100	0.640	0.839	0.082	0.792	0.034	0.530	0.181	0.218	0.738	0.209	0.268
BACE	GCN	IG	scaffold	100	0.850	0.507	0.069	0.800	−0.668	0.140	−0.125	−0.125	0.898	0.067	0.276
BACE[c]	GINE	GNNExpl.	scaffold	30	0.433	0.434	0.189	0.835	−0.128	0.167	0.064	0.214	—	—	—
BACE[c]	GINE	Input×Grad	scaffold	30	0.433	0.434	0.189	0.841	−0.159	0.167	0.079	0.171	—	—	—
BACE	GINE	IG	random	100	0.730	0.846	0.055	0.844	0.307	0.640	0.347	0.278	0.844	0.366	0.219
BACE	GINE	IG	scaffold	100	0.340	0.651	0.179	0.843	**0.389**	0.810	0.104	0.092	0.571	0.184	0.066
BACE[c]	GINE	Saliency	scaffold	30	0.433	0.434	0.189	0.873	−0.174	0.167	0.097	0.173	—	—	—
BBBP	AttentiveFP	IG	random	100	0.770	0.855	0.050	0.750	**0.339**	0.470	0.164	0.167	0.844	0.234	0.203
BBBP	AttentiveFP	IG	scaffold	100	0.950	0.961	0.033	0.841	0.284	0.250	0.036	0.015	0.766	0.082	0.074
BBBP	GAT	IG	random	100	0.780	0.885	0.051	0.770	0.043	0.510	0.328	0.328	0.944	0.105	0.243
BBBP	GAT	IG	scaffold	100	0.980	0.805	0.010	0.831	−0.712	0.000	−0.021	−0.050	0.953	0.021	0.055
BBBP	GCN	IG	random	100	0.790	0.873	0.074	0.768	0.023	0.510	0.311	0.318	0.874	0.122	0.349
BBBP	GCN	IG	scaffold	100	0.980	0.966	0.013	0.814	−0.812	0.030	−0.010	−0.145	0.884	0.065	0.271
BBBP	GINE	GNNExpl.	random	100	0.820	0.886	0.053	0.779	−0.061	0.440	0.287	0.333	0.951	0.097	0.315
BBBP	GINE	GNNExpl.	scaffold	100	0.950	0.932	0.029	0.848	−0.368	0.060	0.018	−0.064	0.791	0.044	0.073
BBBP[c]	GINE	GuidedBP	scaffold	30	0.967	0.805	0.059	0.876	−0.530	0.000	−0.016	−0.037	—	—	—
BBBP[c]	GINE	Input×Grad	scaffold	30	0.967	0.805	0.059	0.839	−0.620	0.000	−0.016	−0.047	—	—	—
BBBP	GINE	IG	random	100	0.820	0.886	0.053	0.724	0.016	0.470	0.332	0.334	0.827	0.098	0.394
BBBP	GINE	IG	scaffold	100	0.950	0.932	0.029	0.841	−0.741	0.040	−0.031	−0.057	0.857	0.028	0.090
BBBP[c]	GINE	PGExpl.	scaffold	30	0.967	0.805	0.059	0.422	−0.185	0.367	−0.055	−0.059	—	—	—
BBBP[c]	GINE	Saliency	scaffold	30	0.967	0.805	0.059	0.826	−0.627	0.000	−0.029	−0.033	—	—	—
BBBP	MPNN	IG	random	100	0.840	0.914	0.053	0.792	0.154	0.420	0.210	0.267	0.765	0.121	0.104
BBBP	MPNN	IG	scaffold	100	0.950	0.922	0.033	0.824	−0.431	0.040	0.052	−0.043	0.832	0.117	0.096
ClinTox	GINE	GNNExpl.	random	100	0.710	0.910	0.157	0.504	0.059	0.430	0.258	0.230	0.984	0.108	0.323
ClinTox	GINE	GNNExpl.	scaffold	100	0.800	0.845	0.062	0.728	−0.242	0.390	0.139	0.135	0.983	0.128	0.289
ClinTox[c]	GINE	Input×Grad	scaffold	30	0.567	0.820	0.207	0.793	0.139	0.567	0.356	0.356	—	—	—
ClinTox	GINE	IG	random	100	0.710	0.910	0.157	0.754	−0.223	0.410	0.230	0.230	0.983	0.108	0.119
ClinTox	GINE	IG	scaffold	100	0.800	0.845	0.062	0.745	−0.329	0.380	0.135	0.135	0.991	0.128	0.254
ClinTox[c]	GINE	Saliency	scaffold	30	0.567	0.820	0.207	0.790	**0.145**	0.567	0.356	0.356	—	—	—

Table 9: Molecular classification cells (no node-level ground truth available), 2 of 2. Every committed classification cell on a real molecular dataset. The ground-truth localisation column does not exist for these datasets — no per-atom labels are published — so only model-side reliability can be measured, which is precisely the gap the Tier-1 cells are needed to close. *charact.* is the GraphFramEx characterisation score and *unfaith.* the PyG/DIG unfaithfulness metric, computed on the same molecules; both are blank for carried rows (^{c}), whose per-molecule records this run did not regenerate. Bold marks the highest occlusion ρ per dataset (emphasis only).

dataset	backbone	attributor	split	n	acc	AUC	ECE	motif top-1	occ. ρ	occ. top-1	Fid+	Fid–	stab.	charact.	unfaith.
DILI ^{c}	GINE	GNNExpl.	scaffold	30	0.800	0.824	0.141	0.826	−0.044	0.433	0.080	0.154	—	—	—
DILI ^{c}	GINE	Input×Grad	scaffold	30	0.800	0.824	0.141	0.858	−0.126	0.433	0.075	0.122	—	—	—
DILI	GINE	IG	random	48	0.688	0.719	0.135	0.760	0.298	0.500	0.354	0.360	0.780	0.383	0.393
DILI	GINE	IG	scaffold	48	0.688	0.824	0.120	0.852	**0.346**	0.521	0.194	0.229	0.941	0.276	0.428
DILI ^{c}	GINE	Saliency	scaffold	30	0.800	0.824	0.141	0.851	−0.128	0.433	0.056	0.134	—	—	—
SIDER	GCN	IG	random	100	0.640	0.677	0.050	0.722	0.479	0.720	0.141	0.249	0.782	0.220	0.048
SIDER	GCN	IG	scaffold	100	0.650	0.663	0.054	0.746	0.067	0.620	0.060	0.118	0.831	0.144	0.062
SIDER ^{c}	GINE	GNNExpl.	scaffold	30	0.633	0.661	0.110	0.816	0.490	0.833	0.038	0.196	—	—	—
SIDER ^{c}	GINE	Input×Grad	scaffold	30	0.633	0.661	0.110	0.878	0.576	0.867	0.088	0.196	—	—	—
SIDER	GINE	IG	random	100	0.710	0.682	0.049	0.761	0.516	0.660	0.157	0.315	0.785	0.221	0.452
SIDER	GINE	IG	scaffold	100	0.570	0.650	0.072	0.836	**0.816**	0.860	0.065	0.082	0.661	0.139	0.490
SIDER ^{c}	GINE	Saliency	scaffold	30	0.633	0.661	0.110	0.887	0.571	0.867	0.093	0.201	—	—	—
Tox21 ^{c}	GINE	GNNExpl.	scaffold	30	1.000	0.827	0.054	0.719	**0.010**	0.367	−0.066	−0.061	—	—	—
Tox21 ^{c}	GINE	Input×Grad	scaffold	30	1.000	0.827	0.054	0.704	−0.409	0.167	−0.075	−0.060	—	—	—
Tox21	GINE	IG	random	100	0.970	0.849	0.099	0.752	−0.739	0.150	−0.058	−0.080	0.639	0.017	0.205
Tox21	GINE	IG	scaffold	100	0.960	0.822	0.028	0.739	−0.367	0.250	−0.011	−0.009	0.381	0.031	0.188
Tox21 ^{c}	GINE	Saliency	scaffold	30	1.000	0.827	0.054	0.697	−0.391	0.167	−0.073	−0.063	—	—	—
hERG ^{c}	GCN	IG	scaffold	30	0.900	0.802	0.098	0.665	**0.711**	0.867	0.589	0.590	—	—	—
hERG ^{c}	GINE	GNNExpl.	scaffold	30	0.733	0.784	0.070	0.725	0.066	0.633	0.094	0.408	—	—	—
hERG ^{c}	GINE	Input×Grad	scaffold	30	0.733	0.784	0.070	0.633	0.342	0.600	0.410	0.408	—	—	—
hERG	GINE	IG	random	66	0.803	0.824	0.093	0.776	0.661	0.833	0.560	0.560	0.992	0.279	0.659
hERG	GINE	IG	scaffold	66	0.727	0.805	0.070	0.689	0.245	0.606	0.337	0.337	0.924	0.283	0.527
hERG ^{c}	GINE	Saliency	scaffold	30	0.733	0.784	0.070	0.626	0.343	0.533	0.409	0.408	—	—	—

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